MODELING SOLID STATE CHEMISTRY
2025

Modelling Structure in Pressure and Mechanical Properties

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MSSC 2025

Overview

Part I: Equation of State and Pressure

How to Model Pressure

Equation of State (EOS)

Input & Output

Modelling Pressure Explicitly (EXTPRESS)

Part II: Elastic Properties

Elastic Properties in CRYSTAL

Input & Output

Overview

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Part I: Equation of State and Pressure

How to Model Pressure

Equation of State (EOS)

Input & Output

Modelling Pressure Explicitly (EXTPRESS)

Part II: Elastic Properties

Elastic Properties in CRYSTAL

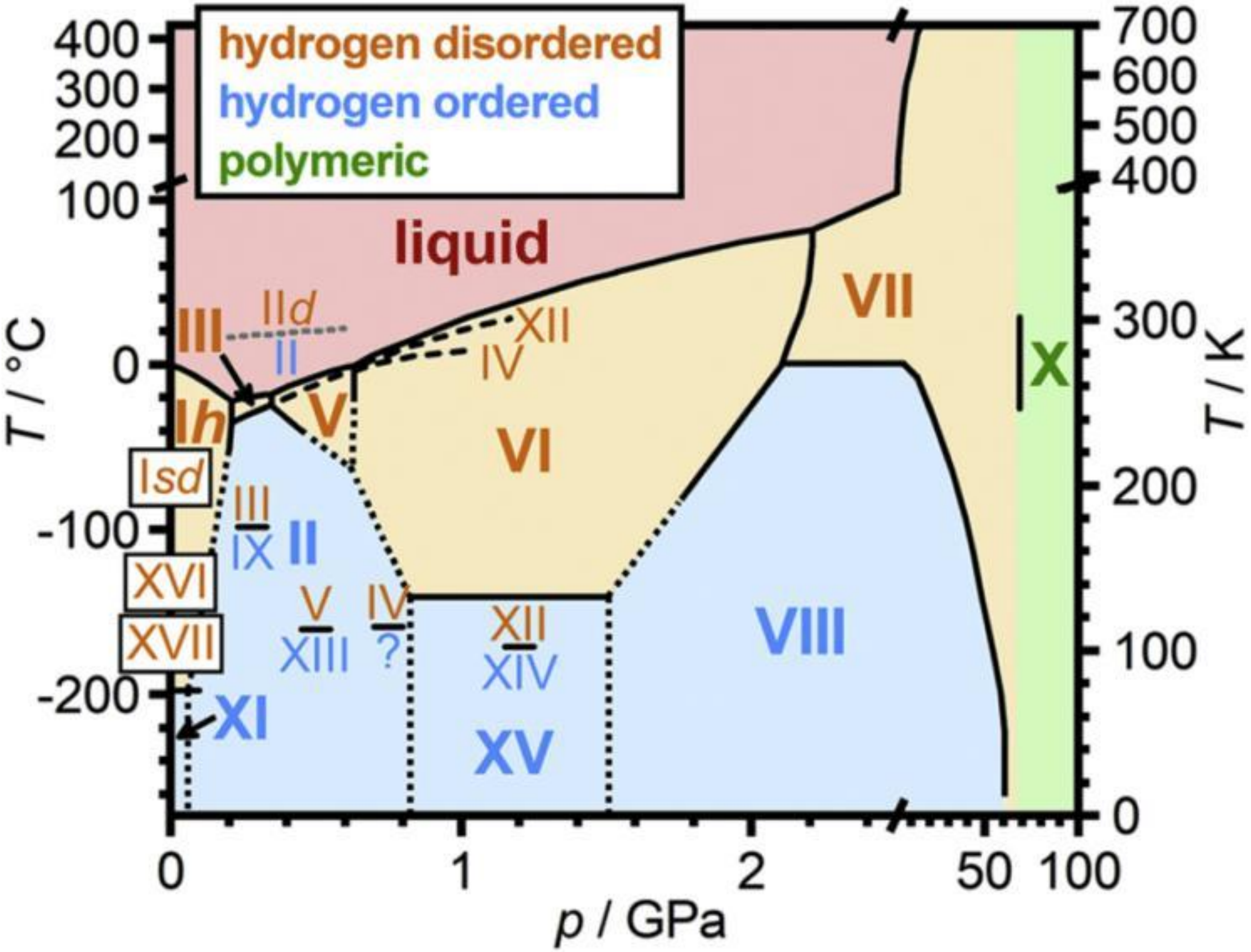
Input & Output

How to Model Pressure

Pressure can greatly affect the lattice, changing the packing, the crystalline structure and inducing phase transintions, but its effect is not simply limited to the crystal structure.

It can alter the electronic structure, induce changes in the magnetic properties, and modify the mechanical response of a material.

How do we model its effect on the lattice?



How to Model Pressure

Constrained Optimizations

Volume Constrained | **Pressure Constrained**

Volume Constrained → **Equation Of State**

Equation of State (EOS)

The Equation of State is a **Pressure-Volume** (or **Energy-Volume**) relationship describing the behavior of a solid under **compression** or **expansion.**

Throughout history several analytical (approximated) expression were proposed. Among these **"Universal" expressions have not been parametrized for specific families of systems.**

It **encompasses information about the bulk modulus and its dependence on pressure**. Additionally, it allows to extrapolate high pressure data starting from low pressure one.

Four different EOS are available in CRYSTAL:

- Murnaghan
- Birch-Murnaghan
- Poirier-Tarantola
- Vinet

Murnaghan

$$E(V) = E_0 + \frac{K_0 V}{B_0'}\left[\left(\frac{V_0}{V}\right)^{K_0'}\frac{1}{K_0'-1}+1\right] - \frac{K_0 V_0}{K_0'-1},$$

$$P(V) = \frac{K_0}{K_0'}\left[\left(\frac{V_0}{V}\right)^{K_0'}-1\right]$$

Birch-Murnaghan

$$E(V) = E_0 + \frac{9V_0K_0}{16}\left\{\left[\left(\frac{V_0}{V}\right)^{\frac{2}{3}}-1\right]^3 K_0' + \left[\left(\frac{V_0}{V}\right)^{\frac{2}{3}}-1\right]^2\left[6-4\left(\frac{V_0}{V}\right)^{\frac{2}{3}}\right]\right\}$$

$$P(V) = \frac{3K_0}{2}\left[\left(\frac{V_0}{V}\right)^{\frac{7}{3}}-\left(\frac{V_0}{V}\right)^{\frac{5}{3}}\right]\left\{1+\frac{3}{4}(K_0'-4)\left[\left(\frac{V_0}{V}\right)^{\frac{2}{3}}-1\right]\right\}$$

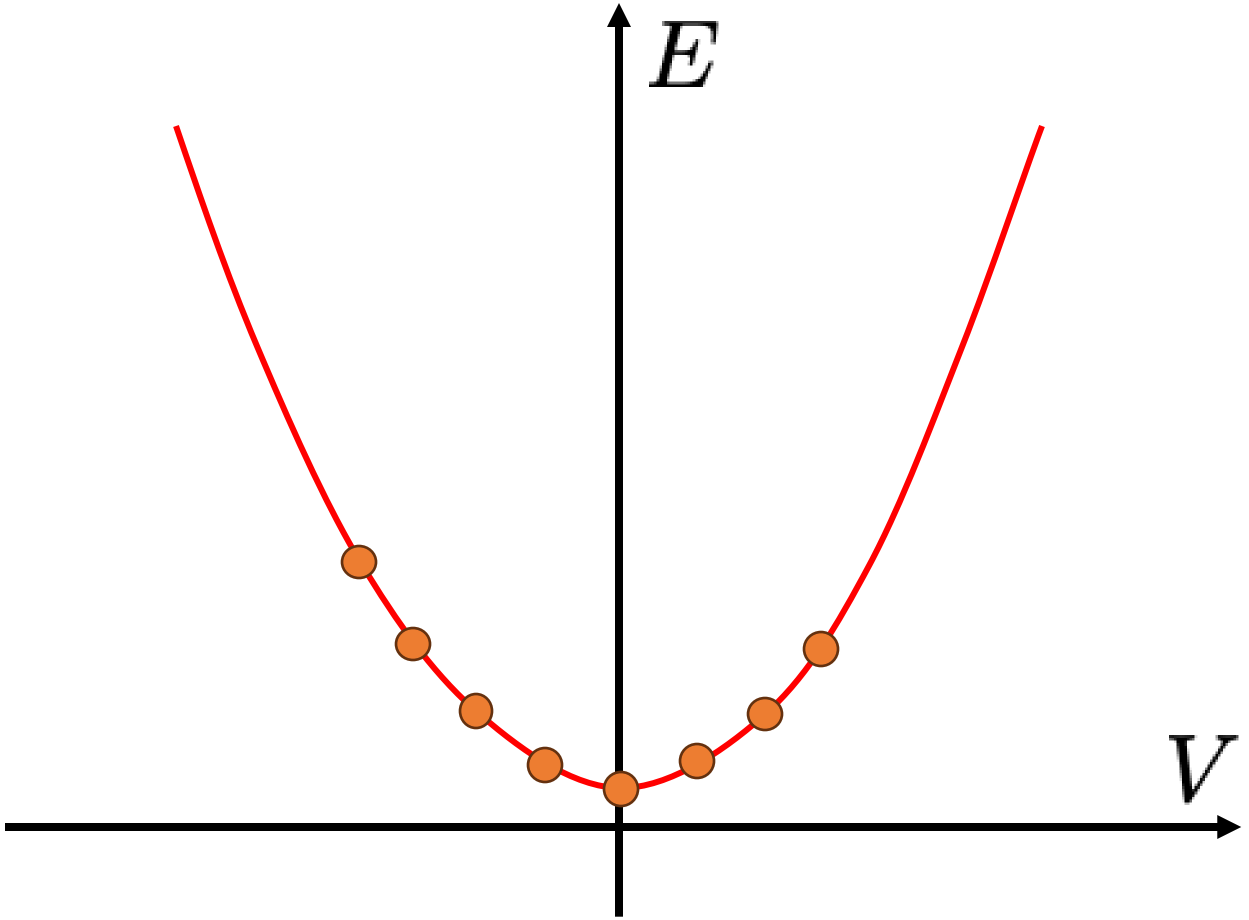
Equation of State (EOS)

A series of volume constrained optimization are performed to obtain the Energy corresponding to the most stable configuration at each volume providing an Energy vs Volume curve.

The obtained volume are then fitted to the EOS:

Murnaghan

$$E(V) = E_0 + \frac{K_0 V}{B_0'}\left[\left(\frac{V_0}{V}\right)^{K_0'}\frac{1}{K_0'-1}+1\right] - \frac{K_0 V_0}{K_0'-1},$$



Starting from the Fit and taking its derivatives, we can derive a set of values

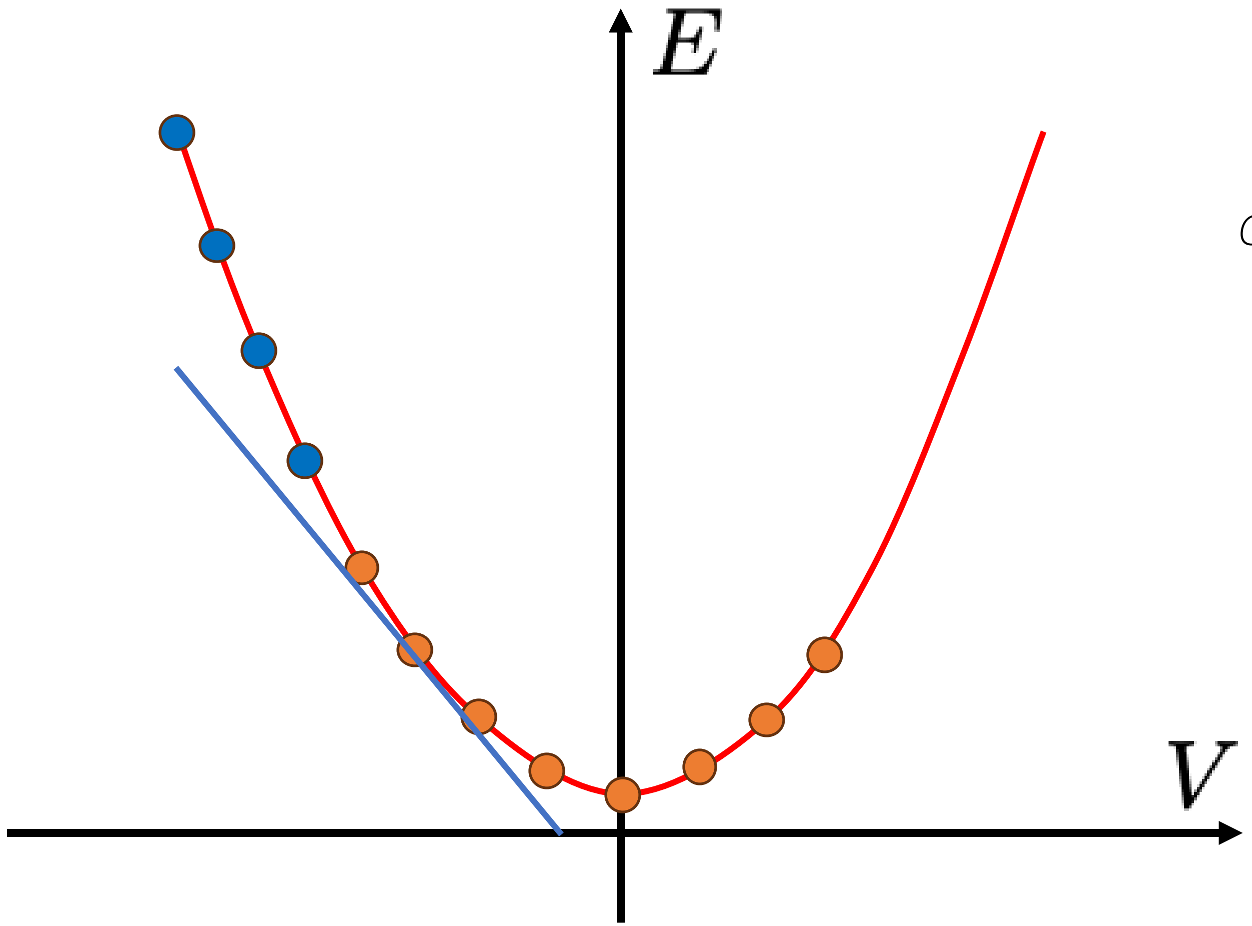
Equation of State (EOS)

Pressure

Taking the derivative if the energy w.r.t. the volume we obtain the pressure at any point of the curve

$$P(V) = -\left(\frac{\partial E}{\partial V}\right)_S$$

$$P(V) = \frac{K_0}{K_0'}\left[\left(\frac{V_0}{V}\right)^{K_0'} - 1\right]$$



Extrapolation

Commonly the computed low-pressure data is used to extrapolate additional high-pressure points

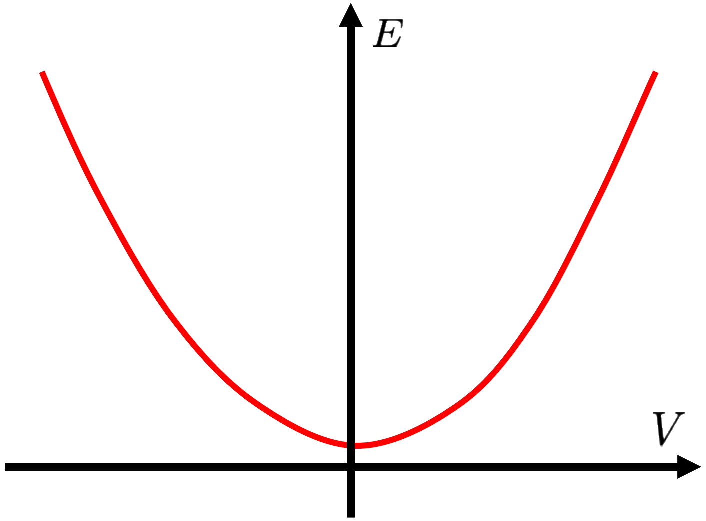
Equation of State (EOS)

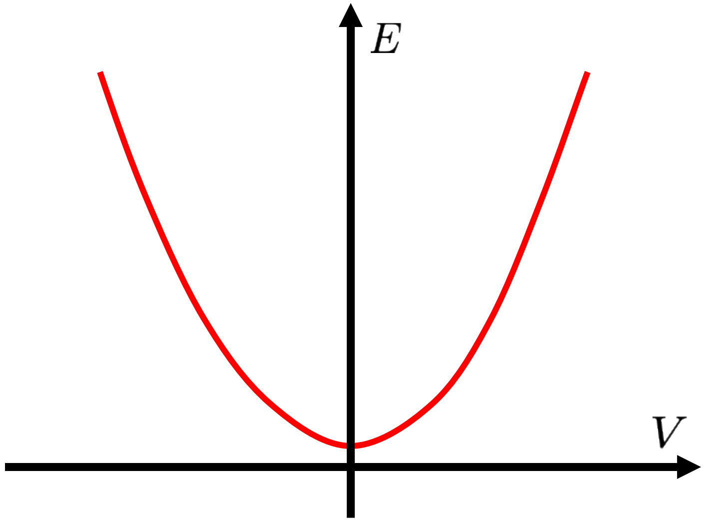
Bulk Modulus

The Bulk Modulus is defined as the curvature of the Energy vs Volume curve. Therefore, taking the second derivative of the energy w.r.t. the volume at any given point of the fit we can extrapolate the bulk modulus at the corresponding volume and pressure.

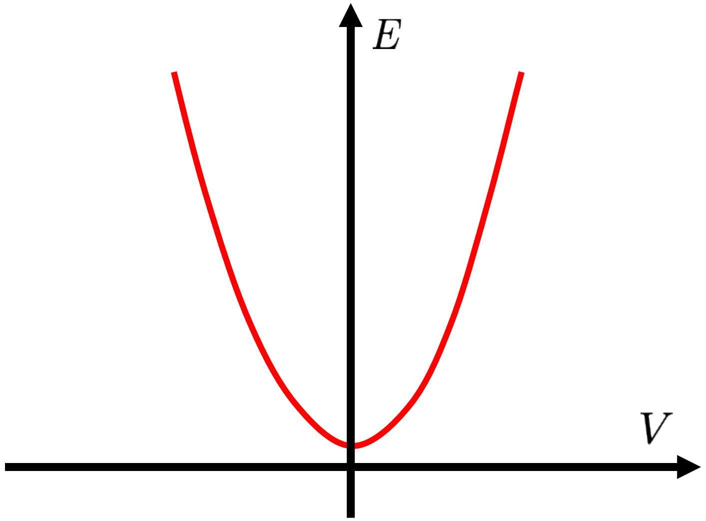
$$K = V\left(\frac{\partial^2 E}{\partial V^2}\right)$$

$K <<$





$K >>$



Input & Output

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Generic Input

. . . Geometry Input . . .

EOS

[Optional Keywords]

END Closes EOS Block

END Closes Geometry Block

-8%

The default calculation will explore six points plus the equilibrium geometry

+8%

```
SORTING VOLUMES/ENERGIES

     VOLUME (A^3)     ENERGY (a.u.)

      63.904327    -1.274250384292E+02
      65.823205    -1.274315735696E+02
      67.975729    -1.274353971252E+02
      69.483386    -1.274361467887E+02
      70.195802    -1.274360050924E+02
      72.487353    -1.274336112279E+02
      74.824376    -1.274285061811E+02
```

The computed points will then be used to perform the fitting with the four available EOS

```
+++++++ FITTING USING ALL POINTS +++++++

 EQUATION OF STATE            VOL(A^3)       E(AU)       BM(GPa)   BM PRIME
 ----------------------------------------------------------------------------
 MURNAGHAN 1944                69.5108    -127.43614405   185.44     4.07
 BIRCH-MURNAGHAN 1947          69.5096    -127.43614568   185.82     4.09
 POIRIER-TARANTOLA 1998        69.5087    -127.43614716   186.17     4.11
 VINET 1987                    69.5091    -127.43614640   185.99     4.10
```

Input & Output

```
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
THERMODYNAMIC FUNCTIONS OBTAINED AT 0 K WITH EOS: BIRCH-MURNAGHAN 1947
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++

V = VOLUME, P = PRESSURE, E = ENERGY, G = GIBBS FREE ENERGY, B = BULK MODULUS

V (A^3)   P (GPa)        E (a.u.)            G (a.u.)       B (GPa)

  63.90     18.56      -127.42503759      -127.15296853      258.85
  64.36     16.75      -127.42687965      -127.17962068      251.93
  64.81     15.00      -127.42853581      -127.20556126      245.21
  65.27     13.31      -127.43001229      -127.23081032      238.68
  65.72     11.67      -127.43131511      -127.25538722      232.33
  66.18     10.09      -127.43245004      -127.27931069      226.16
  66.63      8.56      -127.43342269      -127.30259881      220.16
  67.09      7.08      -127.43423843      -127.32526910      214.32
  67.54      5.65      -127.43490246      -127.34733848      208.64
  68.00      4.27      -127.43541979      -127.36882334      203.11
  68.45      2.93      -127.43579527      -127.38973952      197.74
  68.91      1.64      -127.43603357      -127.41010238      192.51
  69.36      0.39      -127.43613919      -127.42992675      187.42
  69.82     -0.82      -127.43611650      -127.44922703      182.47
  70.27     -1.99      -127.43596969      -127.46801713      177.64
  70.73     -3.12      -127.43570283      -127.48631055      172.95
  71.18     -4.21      -127.43531986      -127.50412036      168.37
  71.64     -5.27      -127.43482455      -127.52145921      163.92
  72.09     -6.30      -127.43422057      -127.53833938      159.58
  72.55     -7.29      -127.43351148      -127.55477277      155.36
  73.00     -8.25      -127.43270069      -127.57077090      151.24
  73.46     -9.17      -127.43179151      -127.58634497      147.23
  73.91    -10.07      -127.43078715      -127.60150580      143.32
  74.37    -10.94      -127.42969071      -127.61626394      139.51
  74.82    -11.78      -127.42850517      -127.63062957      135.80
```

Input & Output

. . . Geometry Input . . .

EOS

[Optional Keywords]

END Closes EOS Block

END Closes Geometry Block

RANGE

VOL1 VOL2 NPOINTS

Allows you to change the default range of volumes and number of point explored in the EOS calculation

Input & Output

```
. . . Geometry Input . . .

EOS
RANGE
0.87 1.02 13
END   Closes EOS Block
END   Closes Geometry Block
```

RANGE

VOL1 VOL2 NPOINTS

Allows you to change the default range of volumes and number of point explored in the EOS calculation

```
SORTING VOLUMES/ENERGIES

    VOLUME (A^3)     ENERGY (a.u.)

     60.437991    -1.274035644427E+02
     61.251885    -1.274096332409E+02
     62.072480    -1.274149953332E+02
     62.890767    -1.274196305873E+02
     63.699420    -1.274235551336E+02
     64.483912    -1.274267743156E+02
     65.298145    -1.274295379009E+02
     66.161298    -1.274318770170E+02
     67.031882    -1.274336307237E+02
     67.909917    -1.274348348685E+02
     68.795529    -1.274355105962E+02
     69.252441    -1.274356585653E+02
     69.701079    -1.274356766717E+02
     70.621281    -1.274353367459E+02
```

The fitting results will change accordingly

```
+++++++ FITTING USING ALL POINTS +++++++

EQUATION OF STATE             VOL(A^3)       E(AU)        BM(GPa)   BM PRIME
----------------------------------------------------------------------------
MURNAGHAN 1944                 69.5383    -127.43568678  187.39      3.76
BIRCH-MURNAGHAN 1947           69.5410    -127.43568546  186.42      3.98
POIRIER-TARANTOLA 1998         69.5432    -127.43568425  185.54      4.18
VINET 1987                     69.5421    -127.43568487  185.99      4.07
```

Input & Output

. . . Geometry Input . . .

EOS

RANGE
0.87 1.02 13

END Closes EOS Block

END Closes Geometry Block

```
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
THERMODYNAMIC FUNCTIONS OBTAINED AT 0 K WITH EOS: BIRCH-MURNAGHAN 1947
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++

V = VOLUME, P = PRESSURE, E = ENERGY, G = GIBBS FREE ENERGY, B = BULK MODULUS

V (A^3)   P (GPa)        E (a.u.)            G (a.u.)        B (GPa)

  60.44     34.58     -127.40356558      -126.92422247      315.34
  60.86     32.40     -127.40682421      -126.95453273      307.58
  61.29     30.29     -127.40987413      -126.98409821      300.03
  61.71     28.24     -127.41272187      -127.01293881      292.68
  62.14     26.26     -127.41537375      -127.04107376      285.53
  62.56     24.34     -127.41783585      -127.06852169      278.56
  62.98     22.48     -127.42011405      -127.09530065      271.78
  63.41     20.68     -127.42221406      -127.12142811      265.17
  63.83     18.93     -127.42414135      -127.14692100      258.74
  64.26     17.24     -127.42590125      -127.17179571      252.47
  64.68     15.60     -127.42749888      -127.19606814      246.36
  65.11     14.01     -127.42893922      -127.21975370      240.41
  65.53     12.47     -127.43022706      -127.24286732      234.61
  65.95     10.97     -127.43136705      -127.26542347      228.95
  66.38      9.52     -127.43236367      -127.28743621      223.44
  66.80      8.11     -127.43322129      -127.30891916      218.06
  67.23      6.75     -127.43394409      -127.32988552      212.82
  67.65      5.43     -127.43453616      -127.35034813      207.71
  68.08      4.14     -127.43500143      -127.37031942      202.72
  68.50      2.90     -127.43534372      -127.38981147      197.86
  68.92      1.69     -127.43556671      -127.40883602      193.12
  69.35      0.52     -127.43567400      -127.42740444      188.49
  69.77     -0.62     -127.43566904      -127.44552780      183.97
  70.20     -1.72     -127.43555520      -127.46321682      179.56
  70.62     -2.79     -127.43533571      -127.48048194      175.26
```

Input & Output

. . . Geometry Input . . .

EOS

[Optional Keywords]

END Closes EOS Block

END Closes Geometry Block

RANGE
VOL1 VOL2 NPOINTS

Allows you to change the default range of volumes and number of point explored in the EOS calculation

RESTART/RESTART2

Allows you to restart an incompleted or completed EOS calculation from the EOSINFO.DAT unit

VRANGE
VMIN VMAX NPOINTS

Allows you to define a custom range of volumes to be printed in the fitting

PRANGE
PMIN PMAX NPOINTS

Allows you to define a custom range of pressures to be printed in the fitting

Input & Output

. . . Geometry Input . . .

EOS

RESTART2

RANGE

0.87 1.02 13

PRANGE

0 35 30

END Closes EOS Block

END Closes Geometry Block

RANGE

VOL1 VOL2 NPOINTS

Allows you to change the default range of volumes and number of point explored in the EOS calculation

RESTART/RESTART2

Allows you to restart an incompleted or completed EOS calculation from the EOSINFO.DAT unit

VRANGE

VMIN VMAX NPOINTS

Allows you to define a custom range of volumes to be printed in the fitting

PRANGE

PMIN PMAX NPOINTS

Allows you to define a custom range of pressures to be printed in the fitting

Will leave unaffected both the Volume Energy Table and the fitting results

Input & Output

. . . Geometry Input . . .

EOS

RESTART2
RANGE
0.87 1.02 13
PRANGE
0 35 30

END Closes EOS Block

END Closes Geometry Block

```
+++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
THERMODYNAMIC FUNCTIONS OBTAINED AT 0 K WITH EOS: BIRCH-MURNAGHAN 1947
+++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++

V = VOLUME, P = PRESSURE, E = ENERGY, G = GIBBS FREE ENERGY, B = BULK MODULUS
```

V (A^3)	P (GPa)	E (a.u.)	G (a.u.)	B (GPa)
69.54	0.00	-127.43568546	-127.43567182	186.43
69.10	1.21	-127.43562438	-127.41648471	191.21
68.67	2.41	-127.43544626	-127.39741454	195.96
68.25	3.62	-127.43515815	-127.37846405	200.69
67.85	4.83	-127.43476643	-127.35962655	205.39
67.45	6.04	-127.43427699	-127.34089952	210.07
67.07	7.24	-127.43369525	-127.32228050	214.72
66.70	8.45	-127.43302614	-127.30376485	219.35
66.34	9.66	-127.43227430	-127.28535079	223.96
65.98	10.86	-127.43144394	-127.26703478	228.55
65.64	12.07	-127.43053914	-127.24881679	233.11
65.30	13.28	-127.42956350	-127.23069262	237.66
64.98	14.48	-127.42852039	-127.21265926	242.19
64.66	15.69	-127.42741320	-127.19471790	246.70
64.34	16.90	-127.42624470	-127.17686294	251.19
64.04	18.10	-127.42501774	-127.15909321	255.67
63.74	19.31	-127.42373489	-127.14140650	260.13
63.45	20.52	-127.42239872	-127.12380256	264.57
63.16	21.72	-127.42101136	-127.10627722	269.00
62.88	22.93	-127.41957517	-127.08883152	273.42
62.61	24.14	-127.41809212	-127.07146265	277.82
62.34	25.35	-127.41656422	-127.05417017	282.20
62.07	26.55	-127.41499302	-127.03694968	286.58
61.81	27.76	-127.41338051	-127.01980256	290.93
61.56	28.97	-127.41172819	-127.00272635	295.28
61.31	30.17	-127.41003778	-126.98572132	299.62
61.07	31.38	-127.40831050	-126.96878378	303.94
60.83	32.59	-127.40654775	-126.95191288	308.25
60.59	33.79	-127.40475083	-126.93510726	312.55
60.36	35.00	-127.40292090	-126.91836507	316.84

Modelling Pressure Explicitly

To determine the Pressure-Volume relationship we use an analytical method based on the stress tensor, here defined as cell gradient w.r.t. the deformed lattice parameter:

$$\sigma_{ij} = \frac{1}{V}\frac{\partial E}{\partial \epsilon_{ij}} = \frac{1}{V}\sum_{k=1}^{3}\frac{\partial E}{\partial a'_{ki}}a_{kj}$$

Deformed lattice parameter

$$a'_{ij} = \sum_{k=1}^{3}(\delta_{jk} + \epsilon_{jk})a_{ik}$$

We can then define a pre-stress in the form of hydrostatic pressure to be summed to the stress tensor

$$\sigma_{ij}^{\text{pre}} = P\delta_{ij}$$

Inverting the first equation we obtain the cell gradient contained by the applied pressure. If we use them in the geometry optimization process, we basically minimize the enthalpy of the system:

$$\frac{\partial H}{\partial a_{ij}} = \frac{\partial E}{\partial a_{ij}} + PV(\mathbf{A}^{-1})_{ji}$$

Modelling Pressure Explicitly

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In practice, on the user side, this translates into the definition of the pressure of choice under the EXTPRESS keyword in the OPTGEOM block. Enabling for the determination of the equilibrium structure at any given pressure to study electronic and vibrational properties.

```
. . . Geometry Input . . .

OPTGEOM

EXTPRESS
0.000679786547

END   Closes OPTGEOM Block

END   Closes Geometry Block
```

Important to Notice!!

The pressure is here defined in atomic unit (*Ha/bohr³*), therefore requiring the user to convert the desired unit in a.u. :

$$1\ Ha/Bohr^3 = 29421\ GPa$$

Modelling Pressure Explicitly

```
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
THERMODYNAMIC FUNCTIONS OBTAINED AT 0 K WITH EOS: BIRCH-MURNAGHAN 1947
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V = VOLUME, P = PRESSURE, E = ENERGY, G = GIBBS FREE ENERGY, B = BULK MODULUS

V (A^3)   P (GPa)        E (a.u.)            G (a.u.)        B (GPa)

  69.54      0.00      -127.43568546      -127.43567182      186.43
  69.10      1.21      -127.43562438      -127.41648471      191.21
  68.67      2.41      -127.43544626      -127.39741454      195.96
  68.25      3.62      -127.43515815      -127.37846405      200.69
  67.85      4.83      -127.43476643      -127.35962655      205.39
  67.45      6.04      -127.43427699      -127.34089952      210.07
  67.07      7.24      -127.43369525      -127.32228050      214.72
  66.70      8.45      -127.43302614      -127.30376485      219.35
  66.34      9.66      -127.43227430      -127.28535079      223.96
  65.98     10.86      -127.43144394      -127.26703478      228.55
  65.64     12.07      -127.43053914      -127.24881679      233.11
  65.30     13.28      -127.42956350      -127.23069262      237.66
  64.98     14.48      -127.42852039      -127.21265926      242.19
  64.66     15.69      -127.42741320      -127.19471790      246.70
  64.34     16.90      -127.42624470      -127.17686294      251.19
  64.04     18.10      -127.42501774      -127.15909321      255.67
  63.74     19.31      -127.42373489      -127.14140650      260.13
  63.45     20.52      -127.42239872      -127.12380256      264.57
  63.16     21.72      -127.42101136      -127.10627722      269.00
  62.88     22.93      -127.41957517      -127.08883152      273.42
  62.61     24.14      -127.41809212      -127.07146265      277.82
  62.34     25.35      -127.41656422      -127.05417017      282.20
  62.07     26.55      -127.41499302      -127.03694968      286.58
  61.81     27.76      -127.41338051      -127.01980256      290.93
  61.56     28.97      -127.41172819      -127.00272635      295.28
  61.31     30.17      -127.41003778      -126.98572132      299.62
  61.07     31.38      -127.40831050      -126.96878378      303.94
  60.83     32.59      -127.40654775      -126.95191288      308.25
  60.59     33.79      -127.40475083      -126.93510726      312.55
  60.36     35.00      -127.40292090      -126.91836507      316.84
```

EOS = 63.45

EXTPRESS = 63.55

```
FINAL OPTIMIZED GEOMETRY - DIMENSIONALITY OF THE SYSTEM      3
(NON PERIODIC DIRECTION: LATTICE PARAMETER FORMALLY SET TO 500)
*******************************************************************************
LATTICE PARAMETERS (ANGSTROMS AND DEGREES) - BOHR = 0.5291772083 ANGSTROM
PRIMITIVE CELL - CENTRING CODE 6/0 VOLUME=    63.546654 - DENSITY 13.372 g/cm^3
        A              B              C           ALPHA      BETA       GAMMA
    6.04906116     6.04906116     6.04906116   147.535214 147.535214  46.571959
*******************************************************************************
ATOMS IN THE ASYMMETRIC UNIT    2 - ATOMS IN THE UNIT CELL:    4
    ATOM                 X/A                 Y/B                 Z/C
*******************************************************************************
     1 T 273 TA    -2.305327346296E-03 -2.305327346296E-03 -4.107065047087E-20
     2 F 273 TA    -2.523053273463E-01  2.476946726537E-01 -5.000000000000E-01
     3 T 233 AS     4.183053273463E-01  4.183053273463E-01 -2.220446049250E-16
     4 F 233 AS     1.683053273463E-01 -3.316946726537E-01 -5.000000000000E-01

TRANSFORMATION MATRIX PRIMITIVE-CRYSTALLOGRAPHIC CELL
 0.0000  1.0000  1.0000  1.0000  0.0000  1.0000  1.0000  1.0000  0.0000

*******************************************************************************
CRYSTALLOGRAPHIC CELL (VOLUME=        127.09330868)
        A              B              C           ALPHA      BETA       GAMMA
    3.38183627     3.38183627    11.11264733    90.000000  90.000000  90.000000

COORDINATES IN THE CRYSTALLOGRAPHIC CELL
    ATOM                 X/A                 Y/B                 Z/C
*******************************************************************************
     1 T 273 TA    -2.289967987061E-20  3.988697520450E-20 -2.305327346296E-03
     2 F 273 TA    -5.000000000000E-01  1.331455080902E-20 -2.523053273463E-01
     3 T 233 AS     5.000000000000E-01  5.000000000000E-01 -8.169467265371E-02
     4 F 233 AS    -1.331455080902E-20 -5.000000000000E-01 -3.316946726537E-01
```

Overview

Part I: Equation of State and Pressure

Part II: Elastic Properties

Overview

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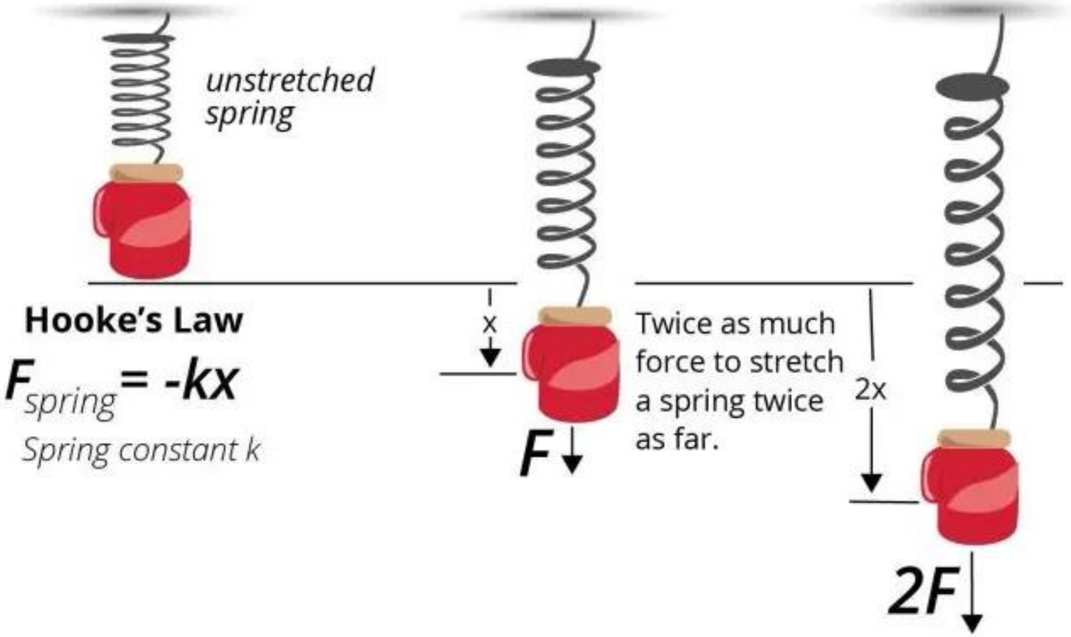
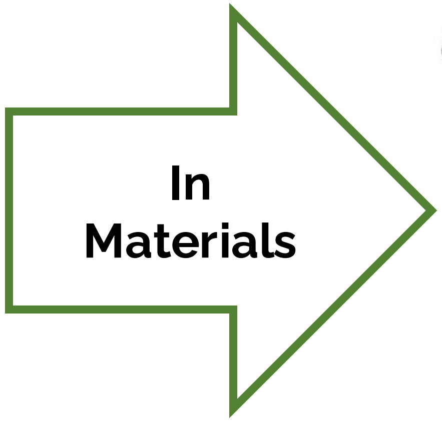
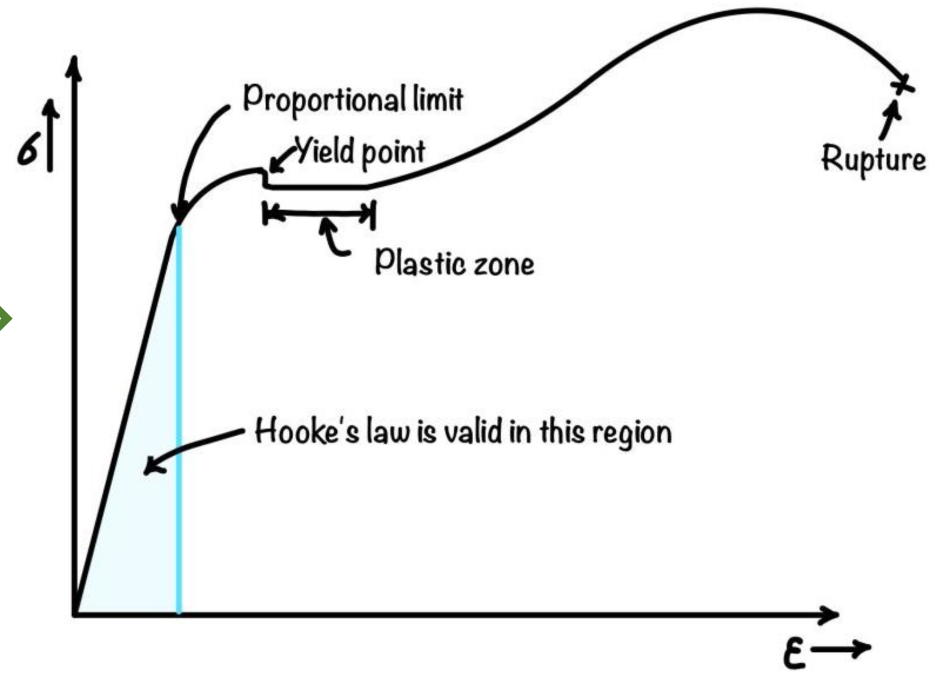
Elastic Properties in CRYSTAL

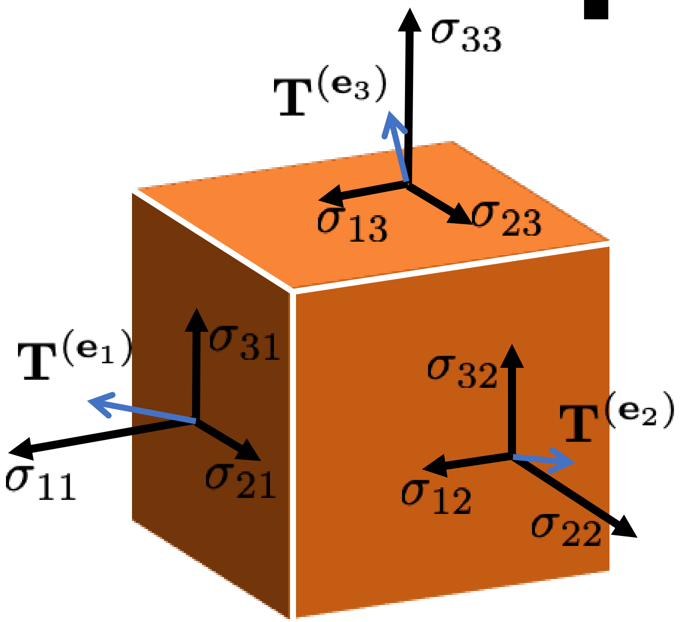






Stress-strain curve for mild steel under uniaxial tension

Elastic Properties in CRYSTAL



Strain tensor in Voigt notation

$$\eta = \begin{array}{cc} \begin{bmatrix} \eta_1 \\ \eta_2 \\ \eta_3 \\ \eta_4 \\ \eta_5 \\ \eta_6 \end{bmatrix} & \begin{array}{c} xx \\ yy \\ zz \\ xy \\ xz \\ yz \end{array} \end{array}$$

Stress tensor

$$\sigma_{ij} = \sum_k \sum_l C_{ijkl}\eta_{kl}$$

I, j, k, l =1,2,3 Cartesian Direction

Elastic constant

$$C_{ijkl} = \frac{1}{V}\left(\frac{\partial^2 E}{\partial \eta_{ij} \partial \eta_{kl}}\right)$$

Generalization of the Hooke's law to an anisotropic media:

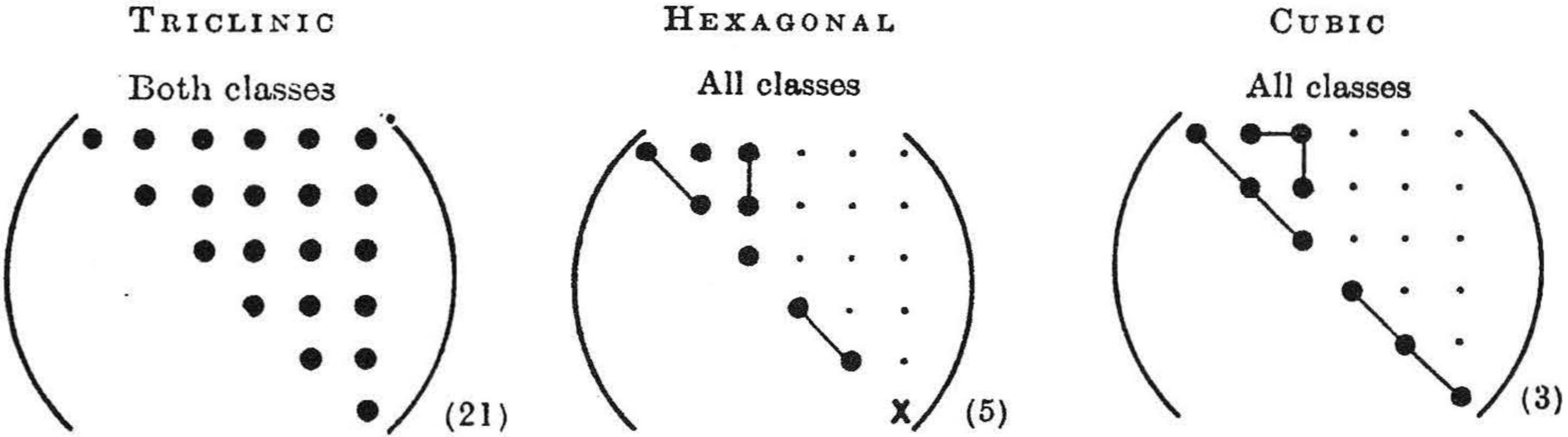
- Stress and Strain Tensor become both second rank tensor
- The elastic constants are a fourth rank tensor acting as the link between the two

81 → **36** → **21**

Stress and Strain tensor are symmetric reducing the number of independent component

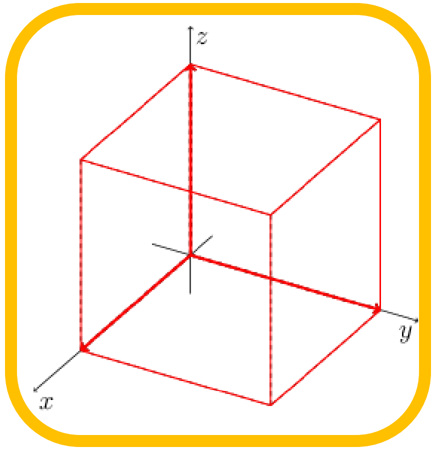
Also C is symmetric further reducing the number of independent element for a triclinic system

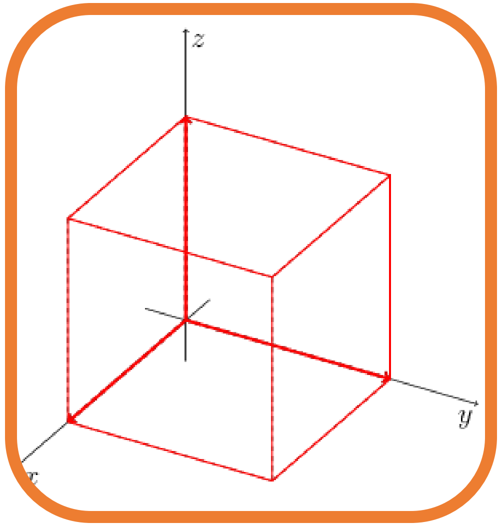
Elastic Properties in CRYSTAL



J. F. Nye, Oxford University Press, (1985)

Elastic Properties in CRYSTAL



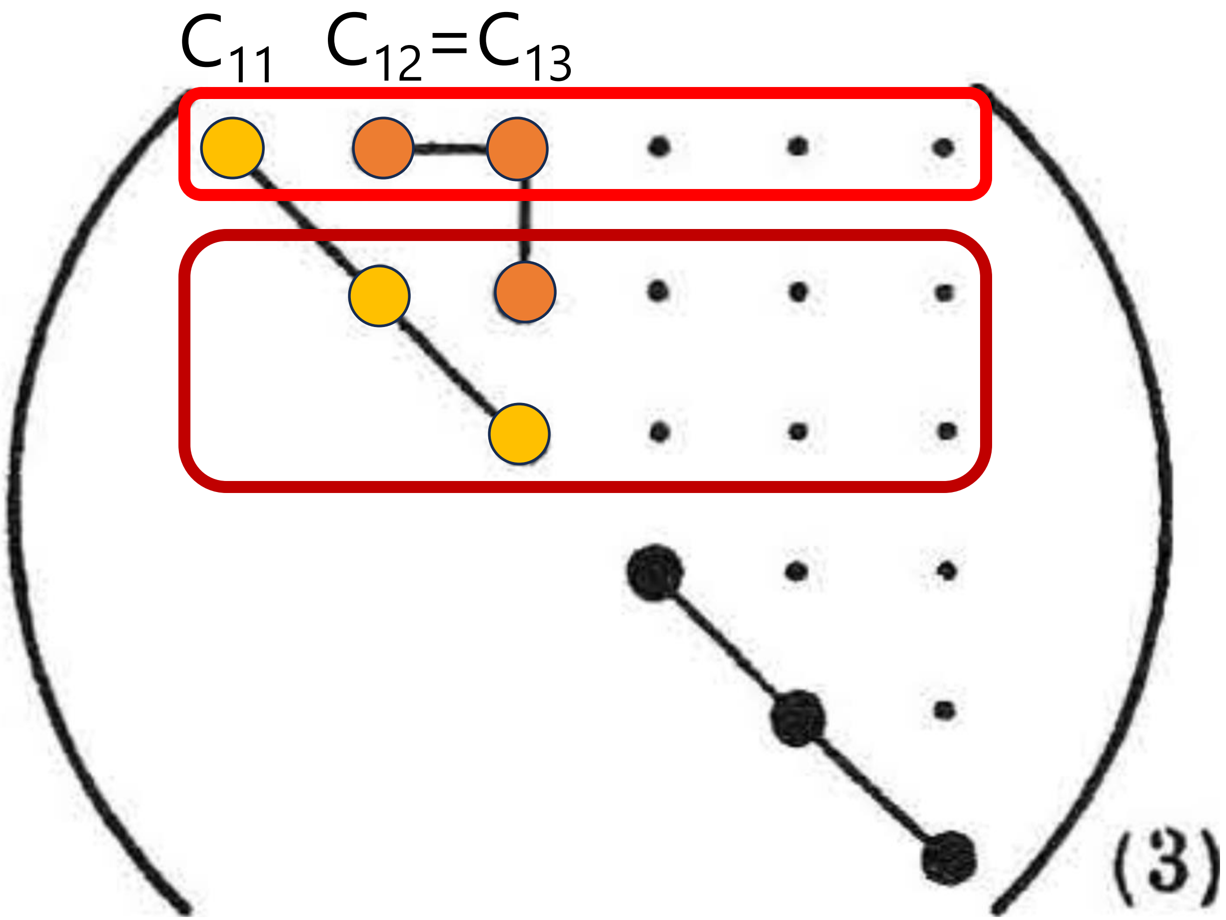


First Deformation

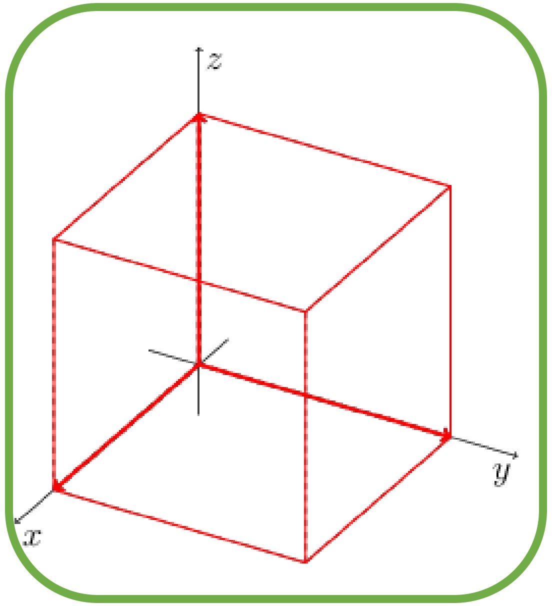
$$C_{11} = C_{22} = C_{33}$$

and

$$C_{13} = C_{23}$$



Elastic Properties in CRYSTAL

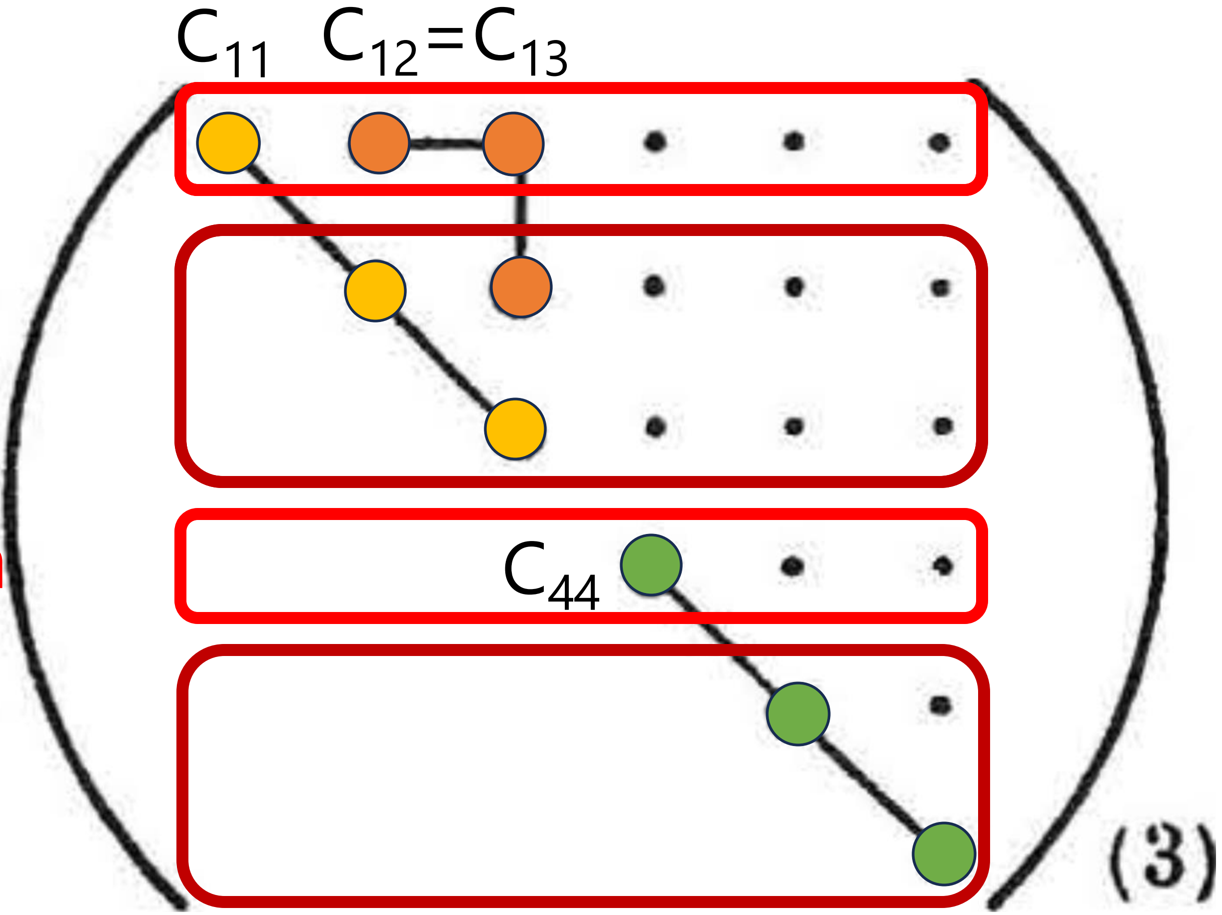


First Deformation

$C_{11} = C_{22} = C_{33}$
and
$C_{13} = C_{23}$

Fourth Deformation

$C_{44} = C_{55} = C_{66}$



Elastic Properties in CRYSTAL

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Starting from the computed elastic (C) and compliance constants ($S=C^{-1}$) it is possible to compute several properties of polycrystalline aggregates via Reuss-Voigt-Hill averaging scheme, here reported for the cubic crystals:

Bulk Modulus

$$K_S = 1/3(C_{11} + 2C_{12}) \equiv 1/3(S_{11} + 2S_{12})^{-1}$$

Shear Modulus

$$\overline{G} = 1/10(C_{11} - C_{12} + 3C_{44}) + 5/2(4(S_{11} - S_{12}) + 3S_{44})^{-1}$$

Young Modulus

$$E = \frac{9K_S\overline{G}}{3K_S + \overline{G}}$$

Poisson's Ratio

$$\sigma = \frac{3K_S - 2\overline{G}}{2(3K_S + \overline{G})}$$

Anisotropy Index

$$A = \frac{2C_{44} + C_{12}}{C_{11}} - 1$$

Additionally, the seismic velocity are computed through through Cristoffel's equation

Elastic Properties in CRYSTAL

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Elastic constants computed at V(P)

$$B_{vu} = \boxed{C_{vu}} + \boxed{\begin{pmatrix} 0 & P & P & 0 & 0 & 0 \\ P & 0 & P & 0 & 0 & 0 \\ P & P & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{-P}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{-P}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{-P}{2} \end{pmatrix}}$$

Eulerian strain correction

If a finite pre-stress tensor is applied in the form of hydrostatic pressure, in the frame of a finite Eulerian strain, it is possible to correct the elastic constants. The *Pressure-Volume* relation, though, needs to be computed previously with either an EOS or an analytical stress tensor approach.

Input & Output

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Generic Input

. . . Geometry Input . . .

ELASTCON

[Optional Keywords]

END Closes ELASTCON Block

END Closes Geometry Block

```
**************************************************************************
**************************************************************************
                          FINAL RESULTS START
**************************************************************************
**************************************************************************

THE CALCULATION HAS BEEN PERFORMED WITH   3 POINTS AND
A STEP OF      0.01000
THIS PERMITS TO PERFORM A FITTING UP TO SECOND   ORDER

DATA FOR MAXIMUM NUMBER OF POINTS AND ORDER OF FIT

SYMMETRIZED ELASTIC CONSTANTS FOR CUBIC        CASE, IN GPa

|    278.419    86.993    86.993     0.000     0.000     0.000 |
|              278.419    86.993     0.000     0.000     0.000 |
|                        278.419     0.000     0.000     0.000 |
|                                  143.781     0.000     0.000 |
|                                            143.781     0.000 |
|                                                      143.781 |


ELASTIC MODULI (COMPLIANCE TENSOR), IN TPa^-1

|     4.2194   -1.0045   -1.0045    0.0000    0.0000    0.0000 |
|               4.2194   -1.0045    0.0000    0.0000    0.0000 |
|                         4.2194    0.0000    0.0000    0.0000 |
|                                   6.9550    0.0000    0.0000 |
|                                             6.9550    0.0000 |
|                                                       6.9550 |
```

Input & Output

Generic Input

. . . Geometry Input . . .

ELASTCON

[Optional Keywords]

END Closes ELASTCON Block

END Closes Geometry Block

```
BULK MODULUS K, SHEAR MODULUS G, YOUNG MODULUS E AND
POISSON RATIO v (ALL IN GPa) ACCORDING TO VOIGT-REUSS-HILL

   K_V      G_V      K_R      G_R      K_H      G_H     E_H       v_H

 150.80   124.55   150.80   119.73   150.80   122.14   288.53    0.181

SEISMIC VELOCITIES BY CHRISTOFFEL EQUATION (km/s)

      WAVE VECTOR                      Vp         Vs1        Vs2

[ 0.000   0.000   1.000]              9.030      6.489      6.489
[ 0.000   1.000   0.000]              9.030      6.489      6.489
[ 1.000   0.000   0.000]              9.030      6.489      6.489
[ 1.000   1.000   0.000]              9.779      6.489      5.295
[ 1.000   0.000   1.000]              9.779      6.489      5.295
[ 0.000   1.000   1.000]              9.779      6.489      5.295
[ 1.000   1.000   1.000]             10.016      5.721      5.721

***************************************************************
***************************************************************
                          FINAL RESULTS END
***************************************************************
***************************************************************


BULK MODULUS (GPa)

K =      150.80
```

Input & Output

. . . Geometry Input . . .

ELASTCON

[Optional Keywords]

END Closes ELASTCON Block

END Closes Geometry Block

NUMDERIV

INUM

Allows you to define the number of points used in the numerical evaluation of the derivatives including the central (*default* = *3*)

STEPSIZE

STEP

Size of the displacement used in the evaluation of the numerical second derivative (*default* = *0.01*)

Input & Output

. . . Geometry Input . . .

ELASTCON

[Optional Keywords]

END Closes ELASTCON Block

END Closes Geometry Block

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INUM

Allows you to define the number of points used in the numerical evaluation of the derivatives including the central (*default* = 3)

STEPSIZE

STEP

Size of the displacement used in the evaluation of the numerical second derivative (*default* = 0.01)

RESTART/RESTART2

Allows you to restart an incompleted or completed ELASTCON calculation from the ELASINFO.DAT unit

Input & Output

```
. . . Geometry Input . . .

ELASTCON
RESTART2
SEISMDIR
3
0 5 1
2 0 1
2 3 4
END   Closes ELASTCON Block

END   Closes Geometry Block
```

NUMDERIV
INUM

Allows you to define the number of points used in the numerical evaluation of the derivatives including the central (*default* = 3)

STEPSIZE
STEP

Size of the displacement used in the evaluation of the numerical second derivative (*default* = 0.01)

RESTART/RESTART2

Allows you to restart an incompleted or completed ELASTCON calculation from the ELASINFO.DAT unit

SEISMDIR
NDIR
DX, DY, DZ

Allows you to define additional direction along which the seismic velocities are computed

Input & Output

. . . Geometry Input . . .

ELASTCON

RESTART2

SEISMDIR

3

0 5 1

2 0 1

2 3 4

END Closes ELASTCON Block

END Closes Geometry Block

```
BULK MODULUS K, SHEAR MODULUS G, YOUNG MODULUS E AND
POISSON RATIO v (ALL IN GPa) ACCORDING TO VOIGT-REUSS-HILL

   K_V      G_V      K_R      G_R      K_H      G_H     E_H       v_H

 150.80   124.55   150.80   119.73   150.80   122.14   288.53    0.181

SEISMIC VELOCITIES BY CHRISTOFFEL EQUATION (km/s)

      WAVE VECTOR                      Vp         Vs1        Vs2

[ 0.000   0.000   1.000]             9.030      6.489      6.489
[ 0.000   1.000   0.000]             9.030      6.489      6.489
[ 1.000   0.000   0.000]             9.030      6.489      6.489
[ 1.000   1.000   0.000]             9.779      6.489      5.295
[ 1.000   0.000   1.000]             9.779      6.489      5.295
[ 0.000   1.000   1.000]             9.779      6.489      5.295
[ 1.000   1.000   1.000]            10.016      5.721      5.721
[ 0.000   5.000   1.000]             9.176      6.489      6.282
[ 2.000   0.000   1.000]             9.557      6.489      5.685
[ 2.000   3.000   4.000]             9.908      6.090      5.523

************************************************************************
************************************************************************
                          FINAL RESULTS END
************************************************************************
************************************************************************


BULK MODULUS (GPa)

K =       150.80
```

Input & Output

```
. . . Geometry Input . . .

ELASTCON

PREOPTGEOM
PRESSURE
5

END   Closes ELASTCON Block

END   Closes Geometry Block
```

```
. . . Geometry Input . . .

ELASTCON

PRESSEOS
PRESSURE
5

END   Closes ELASTCON Block

END   Closes Geometry Block
```